

What Interface Confidence Cannot See: Assembly-Level Steric Validation of De Novo IL-23 p19 Binder Designs

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Abstract—*De novo protein binders are commonly prioritized using confidence scores computed on the designed binder–target complex, but these scores cannot directly evaluate structural partners omitted from the model. We revisit 50 previously accepted miniprotein binders targeting the IL-23 p19 subunit, which were selected using predicted interface confidence, and test whether their designed poses are compatible with intact IL-23. Each design was superposed into the experimental IL-23:IL-23R assembly and evaluated for steric overlap with IL-23R and the obligate p40 subunit. All 50 designs overlap the IL-23R-bound region, but 45 of them also overlap p40, with clash counts ranging from zero to more than 3,500 atom pairs. Using nine interface-confidence metrics, the intended interfaces are distinguished from negative controls, but they do not track p40 overlap. A sequence-only Boltz-2 analysis independently supports the steric result, which is that greater p40 overlap is associated with poorer recovery of the designed binder–p19 interface when p40 is present ($\rho = -0.60$). These results show that interface confidence and assembly compatibility provide distinct information and support assembly-level steric validation as a separate step in de novo binder design.*

Index Terms—de novo protein design, protein–protein interfaces, interface confidence, steric validation, AlphaFold2, Boltz-2, ipSAE, IL-23.

I. INTRODUCTION

Interleukin-23 (IL-23) is a heterodimeric cytokine composed of an IL-23-specific p19 subunit and a shared p40 subunit [1]. Through IL-23R, it promotes pathogenic T helper 17 signaling associated with chronic inflammation [2] and is a well-established therapeutic target in inflammatory bowel disease. p19-directed antibodies such as risankizumab selectively inhibit IL-23 [3], whereas p40-directed agents such as ustekinumab also affect IL-12 [4]. This makes the receptor-facing surface of p19 an attractive selective target [5]. The use of designed miniproteins for this purpose is promising because of their small size and stability [6], and related work has demonstrated oral inhibition of the IL-23 pathway with a designed minibinder [7]. However, p19 does not exist independently in the cytokine: it is joined to p40 by an interchain disulfide, making p40 a fixed part of the local structural environment that any p19-directed binder must accommodate.

Modern de novo binder-design pipelines rank designed candidates after generation. This ranking is often done using predicted interface confidence scores, which can then provide strong evidence that a designed sequence supports the modeled binder–target interface, but they depend on the molecular components included in the prediction. For a target that is a subunit of a larger assembly, a high-confidence interface can

remain geometrically incompatible with a neighboring native partner not included in the modeled complex.

In previous work, we used BindCraft [8] to generate de novo miniprotein binders against this p19 surface [9]. Of 50 accepted designs, Bindcraft generated 42 against isolated p19 and 8 against the intact p19-p40 heterodimer. The designs were selected using AlphaFold2 initial-guess predictions [10] and showed strong predicted interface confidence. The prior screen, however, evaluated confidence in the modeled binder–target interface rather than whether each constructed pose was compatible with the complete IL-23 assembly.

This distinction motivates the present study, in which all 50 original BindCraft poses were placed into the experimental IL-23:IL-23R structure [11] to measure steric overlap with both IL-23R and p40. The analysis then tested whether nine interface-confidence metrics tracked the p40 constraint, with sequence-only Boltz-2 predictions providing an independent check on the steric result. All 50 designed poses overlapped the IL-23R-bound region, but compatibility with p40 varied widely. This variation was not tracked by the interface-confidence scores. However, with Boltz-2, greater p40 overlap was associated with poorer recovery of the designed binder–p19 interface when p40 was present, providing independent support for the assembly-level steric result.

The contributions of this work are:

- An assembly-level steric audit of 50 accepted IL-23 p19 binder designs, separating overlap with the IL-23R-bound region from compatibility with p40.
- An evaluation of nine interface-confidence metrics showing that strong target-interface confidence does not track the p40 steric constraint.
- A sequence-only Boltz-2 analysis providing independent support for the steric result, with greater p40 overlap associated with poorer recovery of the designed binder–p19 interface when p40 is included.

II. RELATED WORK

Computational acceptance of de novo binders. Modern de novo binder design pipelines rely heavily on structure prediction to assess whether a designed sequence supports its intended complex. AlphaFold2 [12] has been adapted for this purpose through AlphaFold initial guess [10], which seeds prediction with a designed pose and tests the interaction afterward. BindCraft [8] incorporates confidence scores from

AlphaFold directly into the generation process, jointly optimizing the binder sequence and interface. Previous IL-23 studies employed these approaches to generate and select 50 miniprotein binders targeting p19 [9]. While these methods provide evidence that the modeled binder–target interface is well supported, they evaluate only the molecular system presented to the predictor. The current study examines whether that interface remains geometrically compatible with the larger biological assembly.

Interface-confidence metrics. Several metrics have been developed to estimate confidence in predicted protein–protein interfaces. AlphaFold-Multimer introduced ipTM [13], while subsequent methods focused more directly on the interacting region. ipSAE [14] scores confidently predicted interchain residue pairs using predicted aligned error and provides several d_0 normalization variants. actipTM [15] weights residues by predicted interface participation, while pDockQ and pDockQ2 [16], [17] combine interface and confidence features to estimate complex quality. The Local Interaction Score (LIS) [18] instead summarizes locally confident regions of the interchain PAE matrix.

These metrics differ in how they define and normalize interface confidence, but all are computed from the molecular system included in the prediction. They can therefore assess confidence in a modeled binder–target interface without directly testing compatibility with a structural partner that is absent from that model. This work evaluates nine such metrics for discrimination and seed reproducibility and separately asks whether any of them track the p40 steric constraint identified at the assembly level.

Assembly context and structural validation. The IL-23 structure makes assembly context especially important. The crystal structure of the cytokine established the p19–p40 heterodimer and its interchain disulfide [1], while the IL-23:IL-23R complex showed how the receptor engages p19 within the intact assembly [11]. A binder designed against isolated p19 can occupy a surface that appears accessible on the subunit, but in the real complex causes conflicts with p40 due to its proximity to p19. Thus, the issue of determining whether a predicted binder pose is geometrically compatible with the complete biological assembly remains even after using interface-confidence metrics to evaluate the designed binders. The present work addresses this by superimposing the designed complexes onto the experimental IL-23 structure, and then measuring steric overlap with both p40 and IL-23R. In this way, assembly-level validation is treated as a separate check from confidence in the designed binder–target interface.

III. METHODS

A. Structures, superposition, and steric overlap. All structural analyses use the IL-23:IL-23R complex PDB 5MZV [11], containing p40 (chain A), p19 (B), IL-23R (C), and crystallization nanobody 22E11 (D). PDB 3DUH [1] represents the unbound cytokine. Residue numbering differs by a constant +19 offset from 3DUH to 5MZV, so the

six design hotspots C51, C80, C81, C89, C137, and C140 correspond to 5MZV residues 70, 99, 100, 108, 156, and 159.

We placed each bindcraft design into the 5MZV frame by superposing its p19 chain onto 5MZV chain B with the Kabsch algorithm [19], as implemented in Biopython [20]. Superposition used 120 sequence-matched anchor residues from the 137 residues resolved in 5MZV chain B (87.6% coverage). The same anchor set applies to all 50 designs because they share the same p19 construct. BindCraft output poses were used directly, avoiding an additional prediction step before steric analysis.

Steric overlap was defined as the number of heavy-atom pairs overlapping by more than 0.4 Å, following the MolProbity serious-clash criterion [21]. Each superposed binder was tested against p40, IL-23R, glycosylated IL-23R, and the crystallization nanobody. The nanobody serves as a negative control because it binds away from the receptor-facing region, while the glycosylated receptor tests whether resolved glycans materially affect the IL-23R overlap measurement.

B. Prediction protocol and counter-screen. All complexes were re-predicted with AlphaFold2 initial guess [10]. The counter-screen contained 200 design–target pairs: 50 cognate on-target complexes, 50 binder–p40 complexes, 50 binder–IL-12 complexes, and 50 matched scrambled-sequence controls. For the on-target arm, the 42 p19-designed binders were paired with isolated p19 and the eight heterodimer-designed binders with intact p19-p40. Each pair was predicted under five independent seeds, yielding 1,000 predictions.

Starting by renumbering the inputs to AlphaFold2 to maintain residue indices across chains, we then placed the binder first, since the initial-guess template assumed a binder-first ordering. After we verified the initial-guess engagement with the target-chain RMSD between the seeded structure and the prediction, we determined that all 1,000 folds passed a 2 Å gate (median 0.42 Å, maximum 1.11 Å). Scrambled controls retained the designed backbone and target but used a shuffled binder sequence, matching the original designs in backbone length and target. By doing this, it provided a matched negative control to assess whether interface confidence depended on the designed sequence instead of just the supplied backbone.

Interfaces were scored with the reference ipSAE implementation [14], [22]. Residue pairs were retained when PAE < 10 Å and C β distance < 15 Å, following the AFDB convention [23]. The primary score was the d_{0res} variant of ipSAE, which uses a residue-based d_0 normalization and was evaluated in the binder→target direction. This is normalized by the number of residues that pass these cutoffs.

C. Statistical analysis and metric evaluation. We computed nine interface-confidence metrics for every AlphaFold2 prediction: ipSAE [14] under its residue-based, chain-based, and domain-based d_0 normalizations; ipTM [13] under the per-chain normalization and recovered directly from the PAE logits; actipTM [15]; pDockQ [16]; pDockQ2 [17]; and the Local Interaction Score (LIS) [18]. We evaluated metric discrimination using AUROC and average precision for the 50 on-target interfaces, the 100 off-target complexes, and the

50 matched scrambled controls. Reproducibility was measured by the within-design range and standard deviation across five prediction seeds.

We measured the association with the assembly-level constraint by the Spearman correlation between each metric and the p40 clash count across the 50 designs, estimating confidence intervals from 5,000 bootstrap resamples, and significance from 10,000 permutations. During robustness analyses, we controlled for binder heavy-atom count and excluded designs with superposition RMSD > 3 Å. We also collapsed repeated MPNN variants into a single median observation per BindCraft backbone. For the nine interface-confidence metrics, we applied a Bonferroni correction to the comparisons ($\alpha = 0.0056$).

Differences between the p19-only and heterodimer design tracks were evaluated using the Mann–Whitney U test [24]. Because the 50 designs derive from 27 BindCraft backbones, we repeated the comparison after collapsing variants within each backbone to their median value.

D. Second-predictor check. All 50 designs were re-predicted with Boltz-2 v2.2.1 [25] using two prediction arms: binder–p19 and binder–p19–p40. Boltz-2 received sequences only, with no starting pose or pocket constraint, and therefore had to locate the interface independently. Since the AlphaFold2 initial guess was seeded with the designed pose, comparisons between the predictors reflect differences in both model and protocol.

Target MSAs were precomputed with ColabFold [26] (263 sequences for p19 and 1,487 for p40), while we supplied de novo binder chains in single-sequence mode. Every design was predicted using five seeds in both arms. The five named leads had 30 two-chain predictions each and 625 Boltz-2 predictions in total (375 two-chain and 250 three-chain). All cross-design and cross-arm comparisons use only the common five seeds so that each design contributes equally. Interfaces were scored using the same ipSAE implementation and the 10/15 Å cutoffs used in the AlphaFold2 analysis.

Pose recovery was evaluated against each design’s original BindCraft complex using CAPRI criteria [27] as implemented in DockQ 2.1 [28]. The primary measure was f_{nat} , the fraction of designed binder–p19 contacts recovered by the prediction, with contacts defined by heavy atoms within 4 Å. f_{nonnat} measured the fraction of predicted contacts absent from the designed interface. DockQ was also recorded, but f_{nat} was emphasized because it directly measures recovery of the designed interface without depending on global structural superposition.

As additional structural controls, we evaluated whether Boltz-2 reconstructed the native p19–p40 interface in the three-chain predictions and recalculated binder–p40 steric overlap directly in each Boltz-2 predicted structure using the same heavy-atom clash criterion as the assembly-level audit. These controls distinguish failure to assemble the cytokine from the binder’s relocation away from its designed pose.

We evaluated the association between p40 clash count and Boltz-2 pose recovery using the same Spearman correlation

and robustness framework described in Section III-C. Associations were evaluated separately for predictions with and without p40.

IV. RESULTS

A. Accepted designs pass original interface-confidence screen

The 50 accepted designs score strongly against their targets under the AlphaFold2 screen. Median on-target ipSAE is 0.738, with 38 of 50 designs reaching the 0.70 confident band [23]. In contrast, none of the 100 off-target complexes reach the 0.60 reliable band (the maximum ipSAE is 0.000 against p40 and 0.013 against intact IL-12). The matched scrambled sequence controls, which retain every designed backbone and replace the binder sequence, also remain at the score floor, with a max ipSAE of 0.000.

The five leads highlighted in the prior study [9] show the same pattern. Their on-target ipSAE values range from 0.776 to 0.804, while all five score 0.000 against p40, IL-12, and their matched scrambled controls (Table I). Thus, by the criteria applied in the original screen, all five appear strongly supported on-target and show no predicted off-target interface with either p40 or IL-12.

TABLE I: Interface-confidence counter-screen for the five leads highlighted in the prior study. Each lead scores well against its intended binder complex but scores below the 0.60 reliable band against p40, intact IL-12, and its matched scrambled-sequence control. Values are median ipSAE across five AlphaFold2 initial-guess seeds, with the seed range in parentheses indicating the on-target predictions.

Lead	ipSAE			
	on-target	p40	IL-12	scrambled
194	0.804 (0.804–0.805)	0.000	0.000	0.000
183	0.801 (0.800–0.801)	0.000	0.000	0.000
1104	0.800 (0.800–0.801)	0.000	0.000	0.000
172	0.796 (0.795–0.797)	0.000	0.000	0.000
192	0.776 (0.775–0.776)	0.000	0.000	0.000

This separation is not specific to ipSAE. Across the nine interface-confidence metrics evaluated on the same predictions, AUROC is 0.999–1.000 against the off-target controls and 1.000 against the matched scrambled controls (Table III). Average precision shows the same behavior, reaching 1.000 for every metric and comparison except LIS against the off-target set (0.998). The scores are also reproducible across the five AlphaFold2 seeds. For ipSAE, the median within-design range is 0.004. The original confidence screen therefore gives a reproducible signal for the interfaces it was designed to evaluate.

B. Assembly-level steric validation reveals a p40 constraint

To evaluate the designed poses with intact IL-23, each design was placed against the experimental IL-23:IL-23R complex (PDB 5MZV) by superposing its p19 chain onto the crystallographic p19 structure [11]. The median RMSD was 1.96 Å, with a range of 1.33–5.01 Å, and 7 of the 50 designs exceeded 3 Å. These values support the use of the superposed

structures for a steric audit and specify how precisely individual clash counts should be interpreted, particularly for the highest-RMSD designs.

All 50 designs overlap the region occupied by IL-23R in the experimental complex. The median was 338 clashing atom pairs involving 16 receptor residues (IQR 226–565). The crystallization nanobody (control), which binds away from the receptor-binding surface, produced no clashes. At the same time, inclusion of the resolved receptor glycans changed the counts for only two designs by at most 18 atom pairs. Receptor overlap is therefore a common property of the design set rather than a feature that distinguishes individual candidates.

The intact assembly reveals a different constraint. Forty-five of the 50 designs place binder atoms inside space occupied by p40, with a median of 412 clashing atom pairs across the full set and values ranging from zero to >3,500 (Fig. 1). This overlap cannot be resolved by dissociation of the partner subunit because p19 and p40 are covalently linked by an interchain disulfide in IL-23 [1], [11]. Thus, a designed pose on p19 must also be geometrically compatible with the space occupied by p40.

The two design tracks show a substantial descriptive difference with respect to this constraint. The 42 designs generated against isolated p19 have a median p40 overlap of 612.5 clashing pairs, compared with 46.5 for the eight designs generated against the intact p19–p40 heterodimer. The heterodimer-designed set also shows greater overlap with the IL-23R-bound region, with median IL-23R clash counts of 1306.5 versus 294.0. These groups are not completely independent. The eight heterodimer designs derive from only four BindCraft backbones, while the 42 p19-only designs derive from 23. After collapsing each backbone to a single observation, the p40 difference remains large in descriptive terms but does not reach statistical significance ($p = 0.0703$), whereas the IL-23R difference remains significant ($p = 0.0014$). The groups also differ in superposition quality, with median RMSDs of 1.71 Å for the heterodimer designs and 2.00 Å for the p19-only designs.

The opposite direction of the IL-23R difference is also notable. The heterodimer-designed set shows less overlap with p40 but the opposite with the IL-23R-bound region. This suggests that avoiding p40 does not simply move designs away from the receptor-facing surface. However, because of the dataset’s unequal group sizes and shared-backbone structure, mechanistic interpretation of this pattern is limited.

The five leads highlighted in the prior study show the same contrast (Fig. 2; Table II). The two heterodimer-designed leads, 183 and 192, show 14 and 48 p40 clashing pairs, respectively. In comparison, the three p19-only leads show substantially greater overlap. However, 172 should be interpreted more cautiously because its superposition RMSD is 5.01 Å, the largest in the set. These examples show directly how designs that occupy the receptor-facing region of p19 can differ greatly in their compatibility with the p40 subunit.

These results separate two properties of the designed poses. Receptor-region overlap is present across the accepted set,

TABLE II: Steric comparison of the five leads. Original rank is the lead ranking reported in the prior study [9]; current rank orders the same five designs by increasing p40 overlap in the intact IL-23 assembly. RMSD is the p19 superposition error onto 5MZV [11]. Clashes are heavy-atom pairs overlapping by more than 0.4 Å (the MolProbity serious-clash criterion) [21]. 172 has the largest superposition error (5.01 Å) and should therefore be interpreted with greater uncertainty.

Rank				RMSD	IL-23R		p40
Original	Steric	Lead	Regime	(Å)	Clashes	Residues	Clashes
2	1	183	full	1.71	574	26	14
5	2	192	full	1.76	1005	47	48
4	3	172	p19-only	5.01	302	13	803
3	4	1104	p19-only	1.90	681	30	1425
1	5	194	p19-only	1.80	396	18	1740

whereas compatibility with p40 varies from essentially no overlap to extensive conflict. Assembly-level compatibility is therefore a separate structural constraint that was not evaluated by the original two-chain interface-confidence screen.

C. Interface confidence does not track the p40 constraint

The steric overlap identified in Section IV-B is not reflected in the interface-confidence score used to select the designs. Across all 50 designs, p40 overlap ranges from 0 to more than 3,500 clashing atom pairs, while on-target ipSAE shows no corresponding decline over that range. The Spearman correlation between p40 overlap and ipSAE is $\rho = +0.15$ (95% CI $[-0.15, +0.43]$, permutation $p = 0.30$), providing no evidence that higher-confidence designs have less steric interference with p40 (Fig. 3).

The lack of a relationship between interface confidence and p40 overlap persists after accounting for several potential sources of bias. Controlling for binder size yields a similar correlation ($\rho = +0.21$); excluding the seven designs with superposition RMSD above 3 Å yields $\rho = +0.12$; and reducing repeated MPNN variants to one observation per BindCraft backbone yields $\rho = +0.18$.

This result is also not specific to ipSAE. If an interface-confidence metric captured assembly compatibility, higher-confidence designs should exhibit fewer p40 clashes, producing a negative association between confidence score and clash burden. Table III tests this expectation across all nine interface-confidence metrics on the same predictions. All nine metrics distinguish the intended binder–target interfaces from the negative controls. However, none of the nine show the expected negative relationship with p40 overlap. Instead, the Spearman correlations range from $\rho = +0.06$ to $+0.32$, and none remain significant after correction for the nine comparisons. Seed-to-seed variation is also small across the metrics, suggesting that unstable confidence estimates do not necessarily account for the absence of a steric signal. 194 provides a clear example, ranking first among all 50 designs by ipSAE in the original study [9], yet it has 1,740 p40 clashing atom pairs. Interface-confidence ranking and assembly-level steric compatibility therefore measure different properties.

IL-23R overlap is universal; p40 overlap varies

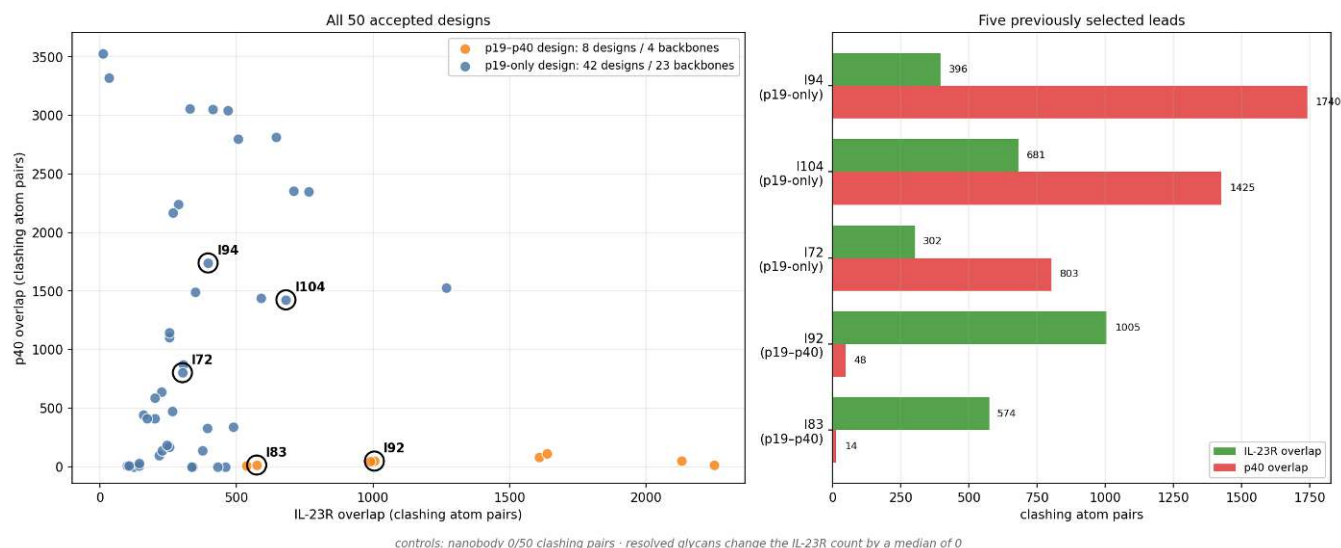


Fig. 1: Assembly-level steric overlap across the 50 accepted designs. **Left:** each design is plotted by its overlap with the space occupied by IL-23R (horizontal) and p40 (vertical) after superposition into the experimental IL-23:IL-23R complex. All 50 designs overlap the IL-23R-bound region, whereas p40 overlap ranges from zero to more than 3,500 clashing atom pairs. Designs are colored by the target used during design; the 42 p19-only designs derive from 23 BindCraft backbones, and the eight p19-p40 designs from four. The five leads highlighted in the prior study are marked separately. **Right:** IL-23R and p40 overlap for those five leads, ordered by increasing p40 overlap. The crystallization nanobody, which binds outside the receptor-facing region, produces zero clashes and serves as a negative control.

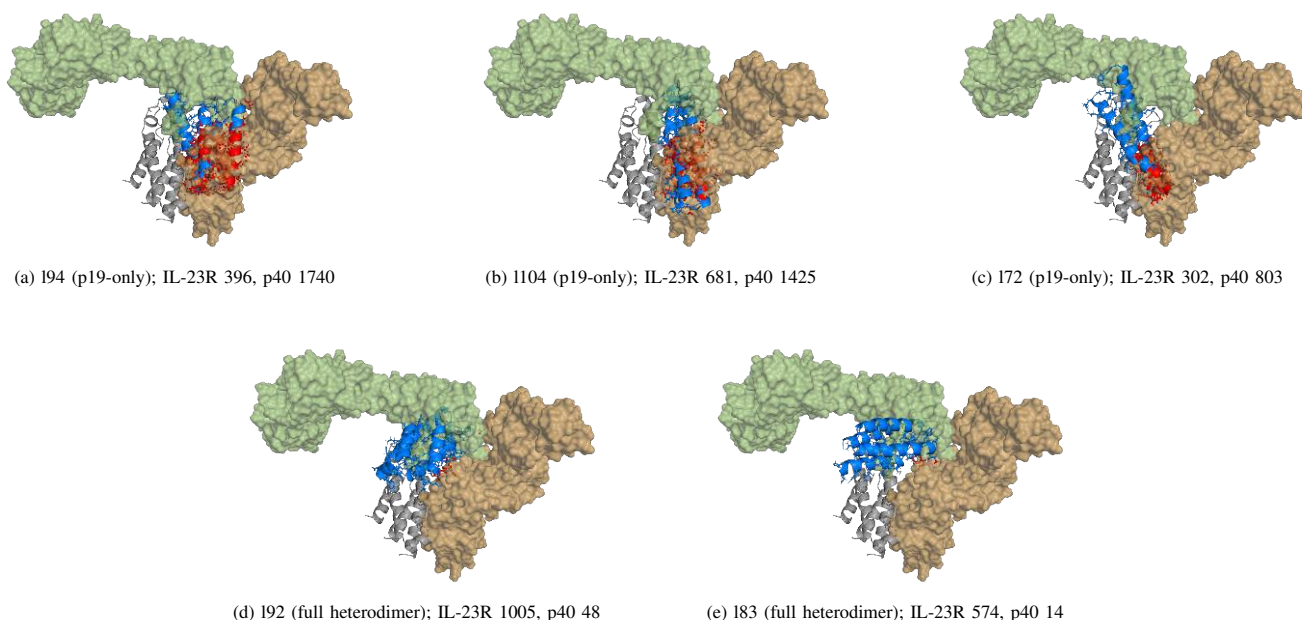


Fig. 2: Assembly-level geometry of the five leads after superposition into the experimental IL-23:IL-23R complex (PDB 5MZV) [11]. In each panel, the binder is **blue** on p19 (**grey**); IL-23R is shown as the **green** surface and p40 as the **tan** surface. **Red** marks binder heavy atoms within 3.5 Å of p40, highlighting close approach to the neighboring subunit. All five leads overlap the IL-23R-bound region, but the three p19-only leads (top row) show extensive p40 overlap, whereas the two heterodimer-designed leads (bottom row) show substantially less. Because p19 and p40 are covalently linked by an interchain disulfide [1], [11], p40 is a fixed part of the local assembly context that the designed pose must accommodate.

D. Boltz-2 predictions support the p40 constraint

We next tested whether the p40 constraint identified by the steric audit appeared under a second structure predictor. Boltz-2 [25] was given sequences only, with no starting pose or pocket constraint, and we predicted each design both with and without p40.

The confidence rankings did not agree between predictors. Across the 50 designs, AlphaFold2 and Boltz-2 ipSAE had only a weak correlation ($\rho = +0.16$). Because this comparison indicates both a change in predictor and in prediction protocol, it does not establish that either score is incorrect. However, this shows that the confidence ranking is not stable across

TABLE III: Interface-confidence metrics distinguish intended interfaces but do not track assembly compatibility. All nine metrics discriminate the 50 on-target designs from the 100 off-target complexes and 50 matched scrambled-sequence controls (AUROC 0.999–1.000). However, if interface confidence reflected compatibility with p40, higher scores would be expected to be associated with fewer clashes ($\rho < 0$), but instead, correlations with p40 clash burden range from $\rho = +0.06$ to $+0.32$. None of which are significant after correction for nine comparisons. Median five-seed ranges are small, indicating that the absence of a steric signal is not explained by prediction-seed variability.

Metric	Interface discrimination (AUROC)		Association with p40 clash burden [†]		Seed reproducibility
	vs. off-target	vs. scrambled	Spearman ρ	95% CI	
ipSAE	1.000	1.000	+0.15	[−0.15, +0.43]	0.004
ipSAE (d0chn)	1.000	1.000	+0.31	[0.00, +0.58]	0.004
ipTM (d0chn)	1.000	1.000	+0.29	[−0.02, +0.57]	0.003
LIS	0.999	1.000	+0.29	[0.00, +0.53]	0.005
ipTM (AF2 exact)	1.000	1.000	+0.10	[−0.19, +0.37]	0.003
actipTM	1.000	1.000	+0.06	[−0.24, +0.36]	0.002
pDockQ	1.000	1.000	+0.31	[+0.02, +0.56]	0.010
pDockQ2	1.000	1.000	+0.13	[−0.18, +0.43]	0.013
ipSAE (d0dom)	1.000	1.000	+0.32	[0.00, +0.58]	0.004

[†] A steric-aware confidence metric would be expected to show a negative association: higher confidence with fewer p40 clashes. None of the nine metrics shows that pattern. Confidence intervals are unadjusted 95% bootstrap intervals; statistical significance was assessed by permutation tests with Bonferroni correction for nine comparisons ($\alpha = 0.0056$), under which none of the metrics is significant.

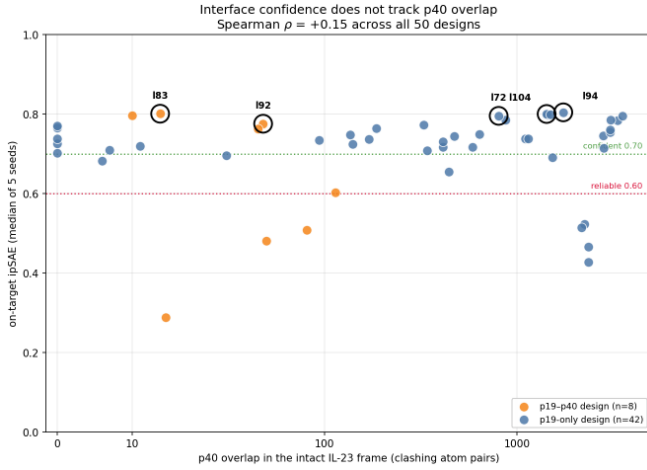


Fig. 3: Interface confidence does not track p40 steric overlap. Each point represents one accepted design, plotted by p40 overlap in the intact IL-23 assembly (horizontal, symmetric log scale) and in on-target ipSAE (vertical, median of five AlphaFold2 initial-guess seeds). Designs are colored by the target used during design, and the five leads highlighted in the prior study are marked separately. p40 overlap spans more than three orders of magnitude without a corresponding decrease in interface confidence (Spearman $\rho = +0.15$, permutation $p = 0.30$).

the two approaches. The sequence-only predictions were also substantially seed-dependent for the five named leads, each of which produced both contacting predictions and predictions at the score floor (Appendix Fig. 5).

The sequence-only predictions provide clearer support for the steric result. With p40 present, designs with greater p40 overlap recover less of their original binder–p19 interface ($\rho = -0.60$; Fig. 4a). The relationship remains even after taking steps to control for binder size ($\rho = -0.59$), excluding designs with superposition RMSD above 3 Å ($\rho = -0.62$), as well as reducing repeated MPNN variants to one observation per BindCraft backbone ($\rho = -0.76$). When p40 is absent, the relationship reverses ($\rho = +0.64$), indicating that the loss of the designed interface is specifically associated with the

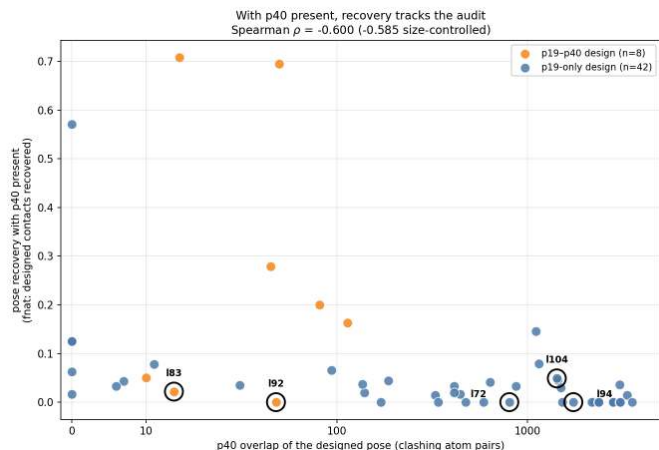
presence of p40. The sign reversal indicates that high p40 clash burden is not intrinsically associated with poor recovery of the designed interface; the negative association emerges specifically when the competing p40 geometry is present. In contrast, AlphaFold2 and Boltz-2 show little agreement in their interface-confidence rankings ($\rho = +0.16$; Fig. 4b).

Two control checks indicate that this effect reflects relocation of the binder rather than a failure to build the cytokine. The native p19–p40 interface assembled in all 250 sequence-only three-chain predictions, and the predicted binder poses showed a median of only 2 clashing atom pairs with p40 in the Boltz-2 frame, compared with the median of 412 measured for the original superposed designed poses. Boltz-2 therefore resolves the steric conflict by moving the binder away from its designed pose.

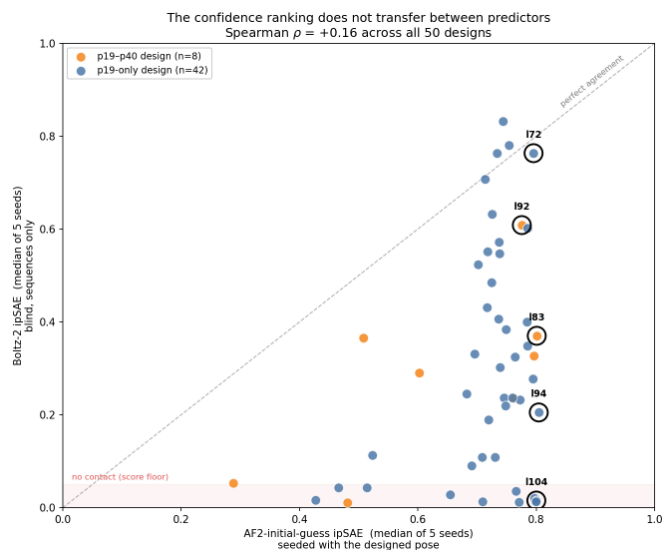
Absolute pose recovery under Boltz-2 is limited (Appendix Fig. 6), so these predictions do not establish that the binders adopt their designed poses or bind experimentally. The result provides an independent check on the steric audit. Namely, designs predicted to interfere most strongly with p40 are also the designs that lose the most of their intended interface when p40 is included. The Boltz-2 results therefore provide an independent check on the assembly-level steric result, whereas the interface-confidence rankings vary between predictors. This observation supports treating assembly-level steric validation as a separate step from the interface-confidence scoring.

V. DISCUSSION

The primary takeaway from this paper is that interface confidence addresses a narrower question than assembly-level validation. The accepted designs show strong and reproducible on-target confidence, yet these scores do not track whether a designed pose conflicts with p40 in intact IL-23. Thus showing that interface confidence and assembly compatibility are distinct properties.



(a) p40 overlap versus recovery of the designed binder-p19 interface under Boltz-2 with p40 present. Greater p40 overlap is associated with lower fnat (Spearman $\rho = -0.60$). Fig. 4: The steric constraint transfers more clearly than the confidence ranking. **Left:** designs with greater p40 overlap recover less of their intended binder-p19 interface when p40 is included in the Boltz-2 prediction. The relationship remains after controlling for binder size and other robustness analyses. **Right:** AlphaFold2 and Boltz-2 ipSAE rankings show little agreement, indicating that confidence ranking is less stable across predictors and protocols than the assembly-level steric effect.



(b) AlphaFold2 and Boltz-2 interface-confidence rankings show only weak agreement across the 50 designs (Spearman $\rho = +0.16$).

This distinction matters when the target is part of a larger biological complex. All 50 designs overlap the IL-23R-bound region, but p40 overlap varies widely. Designs generated against the intact p19-p40 heterodimer show substantially less p40 overlap than those generated against isolated p19. The Boltz-2 analysis provides independent support for the same interpretation. Designs with greater p40 overlap recover less of their intended binder-p19 interface when p40 is included, while this relationship reverses when p40 is absent. By contrast, AlphaFold2 and Boltz-2 show only weak agreement in their interface-confidence rankings.

Together, these findings support treating interface-confidence scoring and assembly-level steric validation as separate steps in computational binder design. They indicate that confidence metrics can evaluate the modeled interface, yet those same metrics cannot directly account for a structure absent from the model that is present in the actual complex. For such targets functioning as assembled complexes, candidate poses should therefore be evaluated in the context of the assembly they are expected to encounter.

This check can also be added without another structure-prediction step. Given the structure of the relevant biological assembly, an existing designed pose can be superimposed onto that assembly and tested directly for overlap with neighboring components. In this dataset, that additional geometric check separated candidates that appeared similarly strong under interface-confidence scoring, including 194, which ranked first by ipSAE but produced 1,740 p40 clashing atom pairs.

Limitations. This study is entirely computational. Steric overlap is measured after rigid superposition onto a single experimental IL-23 structure and therefore does not account for side-chain relaxation or backbone motion. Clash counts also somewhat scale with binder size and depend on su-

perposition quality. We addressed these effects through size-controlled correlations and an RMSD-filtered analysis, but we could not fully eliminate them. Thus, the counts are treated as a continuous measure of geometric conflict. The dataset is also limited to 50 designs against one target. In particular, the eight heterodimer-designed binders derive from only four BindCraft backbones. Because the two design groups were not created under the same conditions and are very unequal in size, the difference between them should be treated as an observed trend rather than as proof that one design strategy is better.

The confidence and second-predictor analyses have additional limitations. The published 0.60 and 0.70 ipSAE interpretation bands were not calibrated specifically for de novo binder design, and an ipSAE value at the score floor does not distinguish a true non-interaction from a prediction in which the chains fail to contact. In essence, absence of evidence does not indicate evidence of absence. Without direct testing, it cannot be determined whether the score reflects a true lack of interaction or a simple failure to make contact in the prediction. Absolute pose recovery under Boltz-2 was also limited, so its results support the relative p40-dependent trend rather than experimental binding or recovery of the designed poses. Finally, AlphaFold2's initial guess and Boltz-2 differ in both their models and prediction protocols, so their confidence rankings cannot be interpreted as a controlled head-to-head comparison of the predictors themselves.

Future work. The next step is experimental validation. Low-p40 overlap and high-p40 overlap candidates should be compared in binding and competition assays, along with functional measurements such as IL-23-dependent STAT3 signaling. This would allow testing for whether the assembly-level steric differences identified here predict actual binding

and receptor inhibition.

Computationally, the steric constraint should be moved earlier in the design process rather than checked only after candidate generation. A future pipeline could include the intact IL-23 assembly during design or add an explicit penalty for clashes with p40. Additional structural predictors may also be useful for testing whether the same trend appears with other modeling strategies. However, experimental testing is needed to determine which geometric thresholds are meaningful for binder performance.

VI. CONCLUSION

This study shows that strong interface confidence does not equate to compatibility with the complete biological assembly. Regarding the 50 accepted IL-23 p19 binder designs, although interface-confidence scores were strong and reproducible, they failed to detect steric conflict with the p40 subunit. Assembly-level superposition revealed substantial variation in p40 compatibility. The sequence-only Boltz-2 analysis provided supporting evidence for the steric interpretation, such that greater p40 overlap associated with a worse recovery of the designed binder-p19 interface when p40 was present. These results support treating interface-confidence scoring and assembly-level steric validation as separate steps in de novo binder design. With targets that exist in complexes, candidate poses should be evaluated in the context of the assembly they are expected to encounter.

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APPENDIX A

SUPPLEMENTARY FIGURES

The supplementary analyses give additional context for the sequence-only Boltz-2 results in Section IV-D.

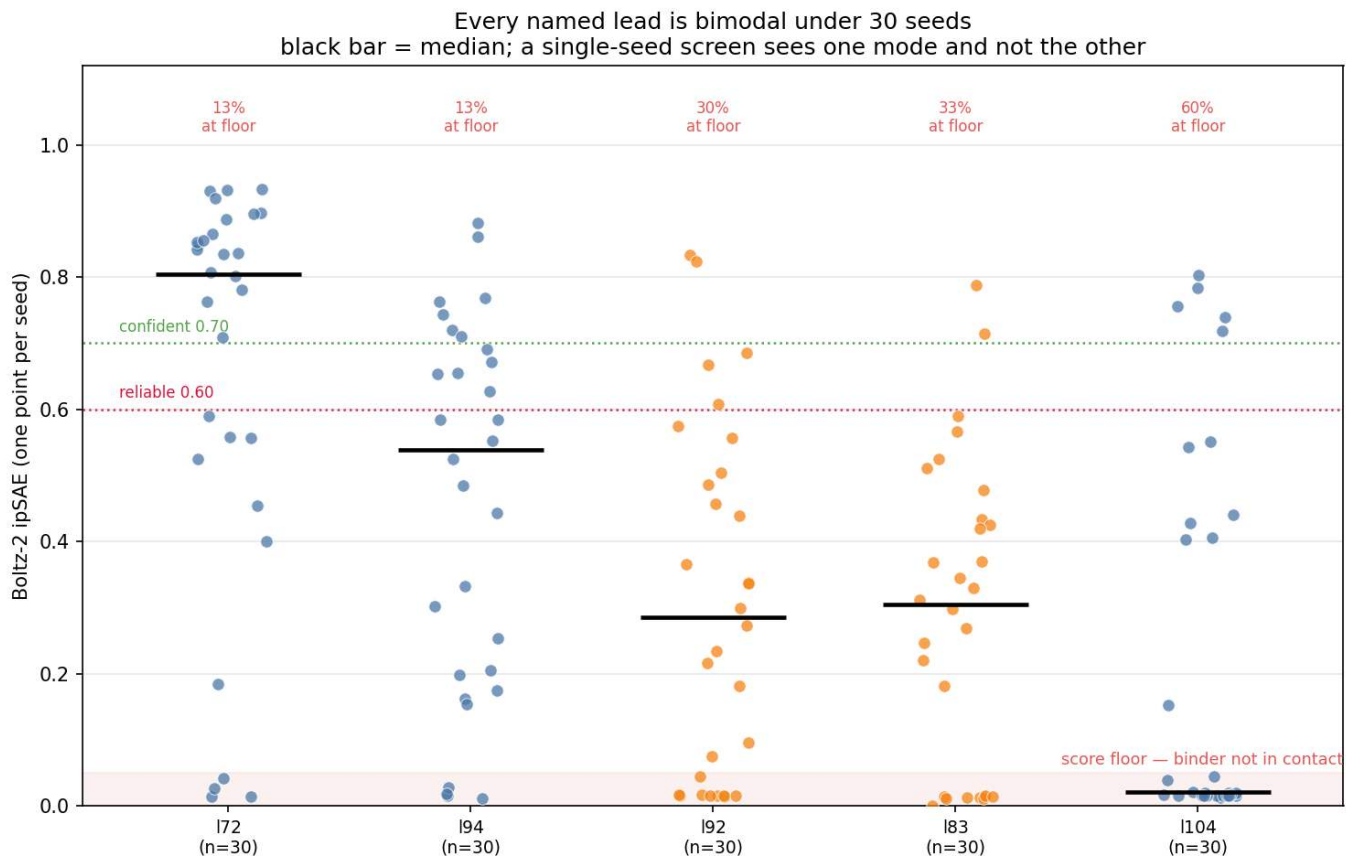


Fig. 5: Seed-to-seed variation for the five named leads under Boltz-2. Each point represents one of 30 sequence-only predictions per lead, and black bars mark the median ipSAE. All five leads show both contacting predictions and predictions at the score floor, where no residue pair satisfies both the distance and PAE cutoffs. Percentages above each lead give the fraction of seeds at the score floor.

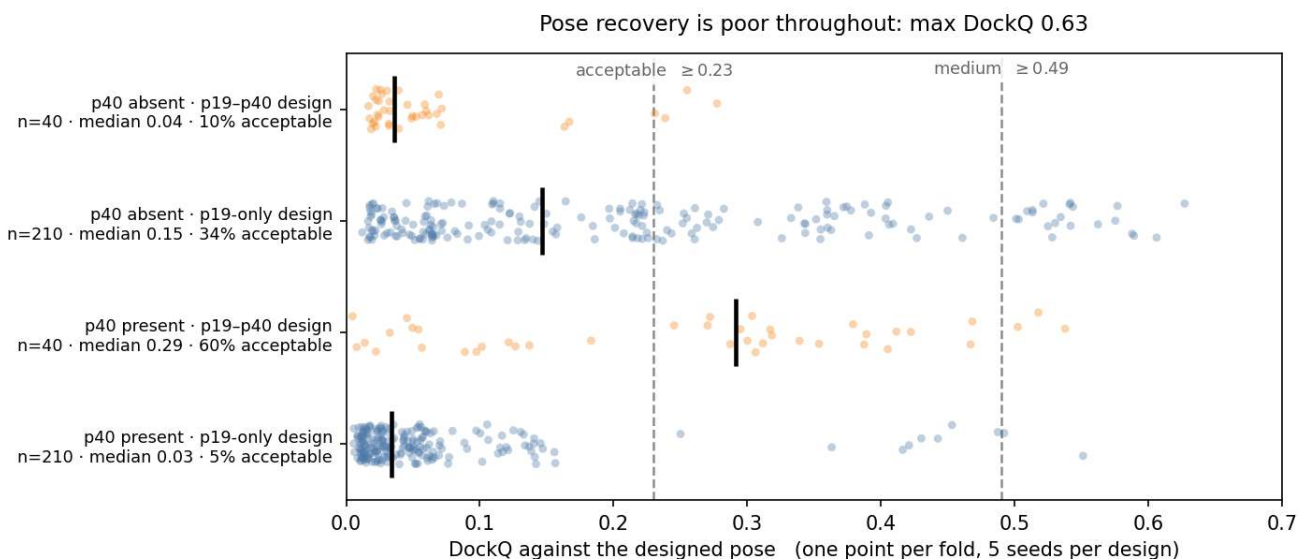


Fig. 6: Pose recovery under Boltz-2 using five balanced seeds per design. Each point is one prediction scored by DockQ against the corresponding BindCraft pose; vertical displacement is jitter only. Black ticks mark group medians, and dashed lines mark the CAPRI acceptable (≥ 0.23) and medium (≥ 0.49) thresholds. Upon p40 being added, the fraction of acceptable predictions increases from 10% to 60% for the heterodimer-designed set and decreases from 34% to 5% for the p19-only set. However, overall absolute pose recovery remains low, so these comparisons reflect relative changes in recovery rather than being a true validation of the designed poses.